

WEEK3:Spring Core and Maven

Exercise 1:Configuring a Basic Spring Application

pom.xml

```
<dependencies>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-context</artifactId>
    <version>5.3.33</version>
  </dependency>
</dependencies>
<build>
  <plugins>
    <plugin>
      <groupId>org.codehaus.mojo</groupId>
      <artifactId>exec-maven-plugin</artifactId>
      <version>3.1.0</version>
      <configuration>
        <mainClass>com.library.LibraryManagementApplication</mainClass>
      </configuration>
    </plugin>
  </plugins>
</build>
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    http://www.springframework.org/schema/beans/spring-beans.xsd">
  <bean id="bookRepository" class="com.library.repository.BookRepository" />
  <bean id="bookService" class="com.library.service.BookService">
    <property name="bookRepository" ref="bookRepository" />
  </bean>
</beans>
```

Repository Class – BookRepository.java

```
package com.library.repository;
public class BookRepository {
    public void saveBook() {
        System.out.println("Book saved to repository.");
    }
}
```

Service Class – BookService.java

```
package com.library.service;
import com.library.repository.BookRepository;
public class BookService {
    private BookRepository bookRepository;
    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }
    public void addBook() {
        System.out.println("Inside BookService.addBook()");
        bookRepository.saveBook();
    }
}
```

LibraryManagementApplication.java

```
package com.library;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;
public class LibraryManagementApplication {
    public static void main(String[] args) {
        ApplicationContext context = new
        ClassPathXmlApplicationContext("applicationContext.xml");
        BookService bookService = context.getBean("bookService",
        BookService.class);
        bookService.addBook();
    }
}
```

Output:

```
LibraryManagement -- zsh -- 80x27
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.518 s
[INFO] Finished at: 2025-07-04T00:45:09+05:30
[INFO] -----
svkeerthi@ss-MacBook-Air LibraryManagement % mvn exec:java -Dexec.mainClass="com
.library.LibraryManagementApplication"

[INFO] Scanning for projects...
[INFO] -----< com.library:LibraryManagement >-----
[INFO] Building LibraryManagement 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.5.1:java (default-cli) @ LibraryManagement ---
Inside BookService.addBook()
Book saved to repository.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.350 s
[INFO] Finished at: 2025-07-04T00:45:19+05:30
[INFO] -----
svkeerthi@ss-MacBook-Air LibraryManagement %
```

Exercise 2: Implementing Dependency Injection

Update **applicationContext.xml**

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="http://www.springframework.org/schema/beans
       http://www.springframework.org/schema/beans/spring-beans.xsd">
  <bean id="bookRepository" class="com.library.repository.BookRepository" />
  <!-- Service Bean with Dependency Injection -->
  <bean id="bookService" class="com.library.service.BookService">
    <property name="bookRepository" ref="bookRepository"/>
  </bean>
</beans>
```

BookService.java

```
package com.library.service;
import com.library.repository.BookRepository;
public class BookService {
    private BookRepository bookRepository;
    public void setBookRepository(BookRepository bookRepository) {
```

```

        this.bookRepository = bookRepository;
    }
    public void addBook() {
        System.out.println("Inside BookService.addBook()");
        bookRepository.saveBook();
    }
}

```

BookRepository.java

```

package com.library.repository;
public class BookRepository {
    public void saveBook() {
        System.out.println("Book saved to repository.");
    }
}

```

LibraryManagementApplication.java

```

package com.library;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;
public class LibraryManagementApplication {
    public static void main(String[] args) {
        ApplicationContext context = new
ClassPathXmlApplicationContext("applicationContext.xml");
        BookService bookService = context.getBean("bookService", BookService.class);
        bookService.addBook();
    }
}

```

Output:

```

[INFO] Compiling 4 source files with javac [debug target 1.8] to target/classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
    not setting the bootstrap class path may lead to class files that cannot run on JDK 8
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap cla
ss path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.516 s
[INFO] Finished at: 2025-07-04T00:57:28+05:30
[INFO] -----
[INFO] Scanning for projects...
[INFO] -----< com.library:LibraryManagement >-----
[INFO] Building LibraryManagement 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.5.1:java (default-cli) @ LibraryManagement ---
Inside BookService.addBook()
Book saved to repository.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.330 s
[INFO] Finished at: 2025-07-04T00:57:29+05:30
[INFO] -----
svkeerthi@ss-MacBook-Air LibraryManagement %

```

Exercise 4: Creating and Configuring a Maven Project

Pom.xml

```
<dependencies>
  <!-- Spring Core -->
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-context</artifactId>
    <version>5.3.33</version>
  </dependency>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-aop</artifactId>
    <version>5.3.33</version>
  </dependency>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-webmvc</artifactId>
    <version>5.3.33</version>
  </dependency>
  <dependency>
    <groupId>junit</groupId>
    <artifactId>junit</artifactId>
    <version>4.13.2</version>
    <scope>test</scope>
  </dependency>
</dependencies>
<build>
  <plugins>
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-compiler-plugin</artifactId>
      <version>3.11.0</version>
      <configuration>
        <source>1.8</source>
        <target>1.8</target>
      </configuration>
    </plugin>
    <plugin>
      <groupId>org.codehaus.mojo</groupId>
      <artifactId>exec-maven-plugin</artifactId>
      <version>3.1.0</version>
      <configuration>
        <mainClass>com.library.LibraryManagementApplication</mainClass>
      </configuration>
    </plugin>
  </plugins>
</build>
```


Output:

```
LibraryManagement -- zsh -- 80x27
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ LibraryManagement ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ LibraryManagement ---
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ LibraryManagement ---
[INFO] Building jar: /Users/svkeerthi/LibraryManagement/target/LibraryManagement-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- install:3.1.2:install (default-install) @ LibraryManagement ---
[INFO] Installing /Users/svkeerthi/LibraryManagement/pom.xml to /Users/svkeerthi/.m2/repository/com/library/LibraryManagement/1.0-SNAPSHOT/LibraryManagement-1.0-SNAPSHOT.pom
[INFO] Installing /Users/svkeerthi/LibraryManagement/target/LibraryManagement-1.0-SNAPSHOT.jar to /Users/svkeerthi/.m2/repository/com/library/LibraryManagement/1.0-SNAPSHOT/LibraryManagement-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.812 s
[INFO] Finished at: 2025-07-04T01:16:03+05:30
[INFO] -----
svkeerthi@ss-MacBook-Air LibraryManagement %
```

Exercise 5:Configuring the Spring IoC Container

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           http://www.springframework.org/schema/beans/spring-beans.xsd">
    <bean id="bookRepository" class="com.library.repository.BookRepository" />
    <bean id="bookService" class="com.library.service.BookService">
        <property name="bookRepository" ref="bookRepository"/>
    </bean>
</beans>

package com.library.repository;

public class BookRepository {
    public void saveBook() {
        System.out.println("Book saved to repository.");
    }
}
```

```

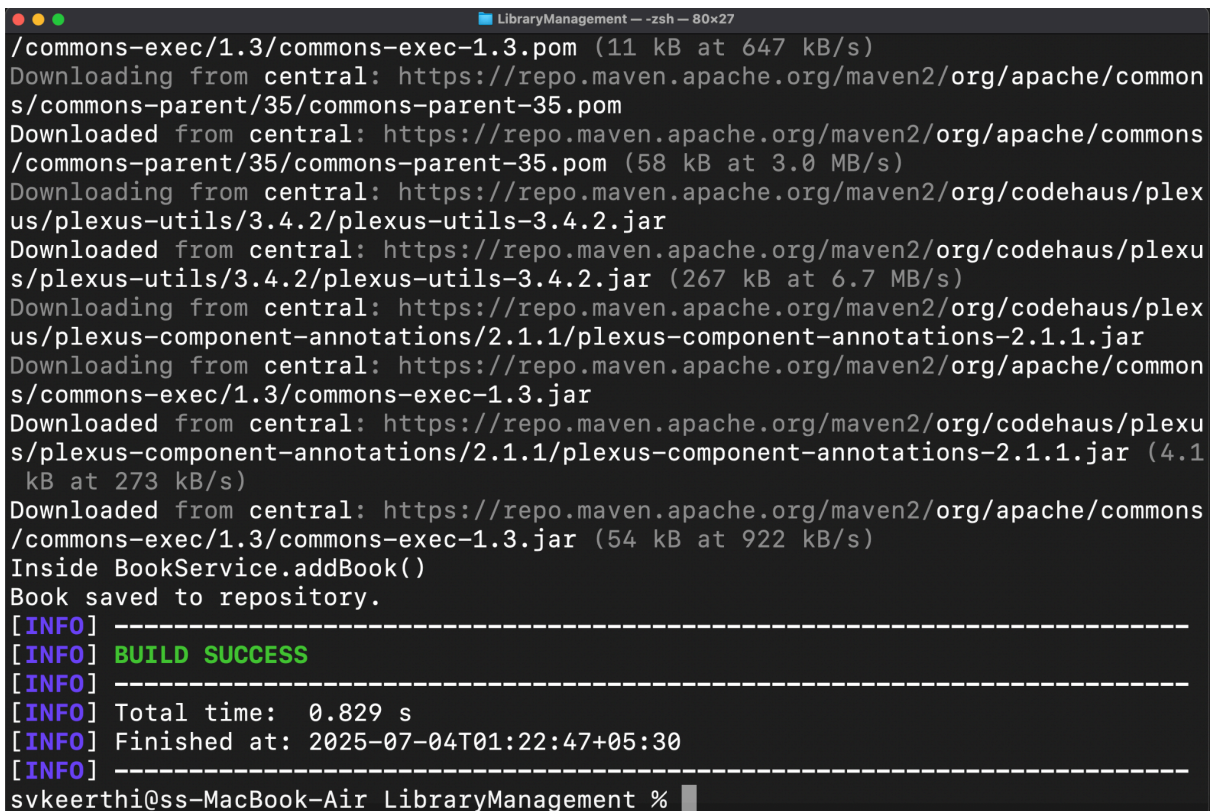
package com.library.service;
import com.library.repository.BookRepository;
public class BookService {

    private BookRepository bookRepository;
    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }
    public void addBook() {
        System.out.println("Inside BookService.addBook()");
        bookRepository.saveBook();
    }
}

package com.library;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;
public class LibraryManagementApplication {
    public static void main(String[] args) {
        ApplicationContext context = new
ClassPathXmlApplicationContext("applicationContext.xml");
        BookService bookService = context.getBean("bookService", BookService.class);
        bookService.addBook();
    }
}

```

Output:



```

LibraryManagement - zsh - 80x27
/commons-exec/1.3/commons-exec-1.3.pom (11 kB at 647 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/common
s/commons-parent/35/commons-parent-35.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons
/commons-parent/35/commons-parent-35.pom (58 kB at 3.0 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plex
us/plexus-utils/3.4.2/plexus-utils-3.4.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexu
s/plexus-utils/3.4.2/plexus-utils-3.4.2.jar (267 kB at 6.7 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plex
us/plexus-component-annotations/2.1.1/plexus-component-annotations-2.1.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/common
s/commons-exec/1.3/commons-exec-1.3.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexu
s/plexus-component-annotations/2.1.1/plexus-component-annotations-2.1.1.jar (4.1
kB at 273 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons
/commons-exec/1.3/commons-exec-1.3.jar (54 kB at 922 kB/s)
Inside BookService.addBook()
Book saved to repository.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.829 s
[INFO] Finished at: 2025-07-04T01:22:47+05:30
[INFO] -----
svkeerthi@ss-MacBook-Air LibraryManagement %

```

Exercise 7: Implementing Constructor and Setter Injection

Update **BookService**

```
package com.library.service;
import com.library.repository.BookRepository;
public class BookService {

    private BookRepository bookRepository;
    private String serviceName;
    public BookService(String serviceName) {
        this.serviceName = serviceName;
    }
    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }
    public void addBook() {
        System.out.println "[" + serviceName + "] Inside BookService.addBook()");
        bookRepository.saveBook();
    }
}
package com.library.repository;

public class BookRepository {
    public void saveBook() {
        System.out.println("Book saved to repository.");
    }
}
```

Update **applicationContext.xml**

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           http://www.springframework.org/schema/beans/spring-beans.xsd">
    <bean id="bookRepository" class="com.library.repository.BookRepository" />
    <bean id="bookService" class="com.library.service.BookService">
        <!-- Constructor argument: string -->
        <constructor-arg value="Library Management Service" />
        <property name="bookRepository" ref="bookRepository" />
    </bean>
</beans>
```

LibraryManagementApplication.java

```
package com.library;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;
```



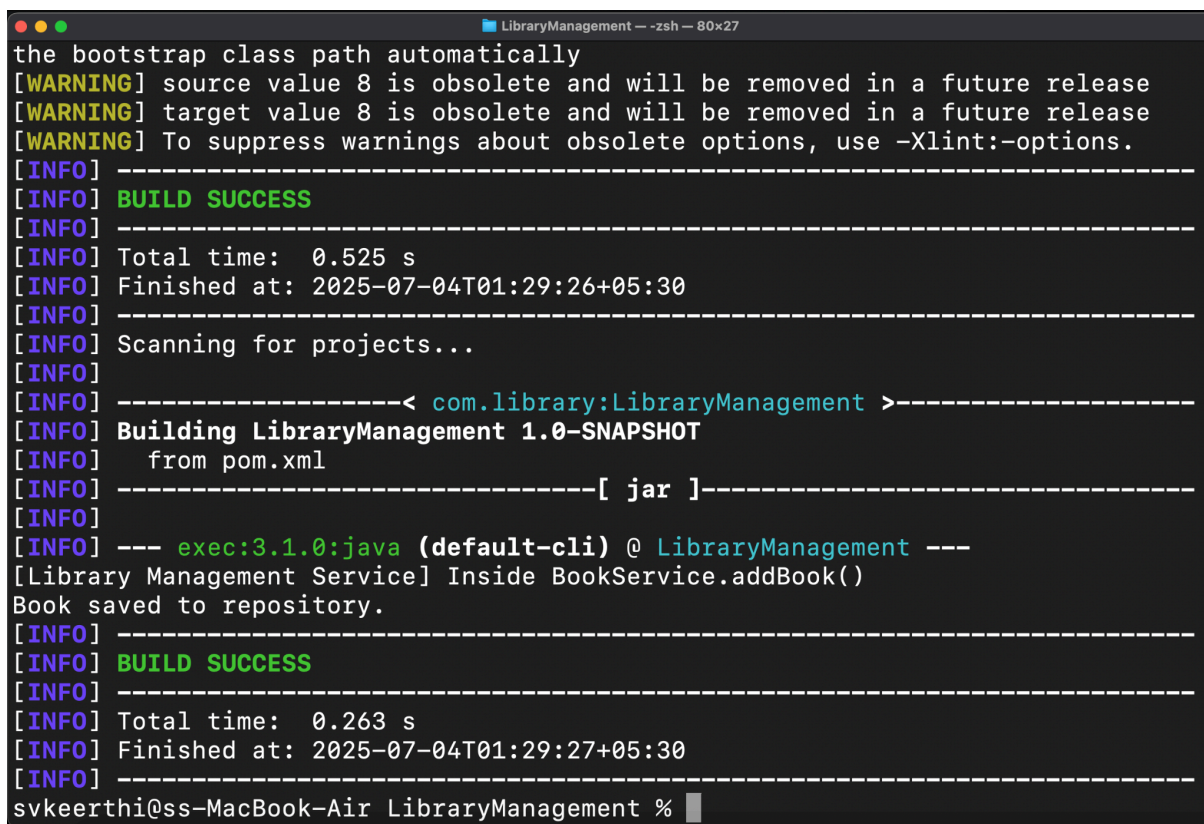
```

public class LibraryManagementApplication {
    public static void main(String[] args) {
        ApplicationContext context = new
        ClassPathXmlApplicationContext("applicationContext.xml");

        BookService bookService = context.getBean("bookService", BookService.class);
        bookService.addBook();
    }
}

```

Output:



```

the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.525 s
[INFO] Finished at: 2025-07-04T01:29:26+05:30
[INFO] -----
[INFO] Scanning for projects...
[INFO] -----< com.library:LibraryManagement >-----
[INFO] Building LibraryManagement 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.1.0:java (default-cli) @ LibraryManagement ---
[Library Management Service] Inside BookService.addBook()
Book saved to repository.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.263 s
[INFO] Finished at: 2025-07-04T01:29:27+05:30
[INFO] -----
svkeerthi@ss-MacBook-Air LibraryManagement %

```

Exercise 9: Creating a Spring Boot Application

```
spring.datasource.url=jdbc:h2:mem:librarydb
spring.datasource.driverClassName=org.h2.Driver
spring.datasource.username=sa
spring.datasource.password=
```

```
spring.jpa.database-platform=org.hibernate.dialect.H2Dialect
spring.h2.console.enabled=true
spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update
```

```
package com.library.model;
```

```
import jakarta.persistence.*;
```

```
@Entity
```

```
public class Book {
```

```
    @Id
```

```
    @GeneratedValue(strategy = GenerationType.IDENTITY)
```

```
    private Long id;
```

```
    private String title;
```

```
    private String author;
```

```
    // Constructors
```

```
    public Book() {}
```

```
    public Book(String title, String author) {
```

```
        this.title = title;
```

```
        this.author = author;
```

```
    }
```

```
    public Long getId() { return id; }
```

```
    public void setId(Long id) { this.id = id; }
```

```
    public String getTitle() { return title; }
```

```
    public void setTitle(String title) { this.title = title; }
```

```
    public String getAuthor() { return author; }
```

```
    public void setAuthor(String author) { this.author = author; }
```

```
}
```

```
package com.library.repository;
```

```
import com.library.model.Book;
```

```
import org.springframework.data.jpa.repository.JpaRepository;
```

```
public interface BookRepository extends JpaRepository<Book, Long> {
```

```
}
```

```

package com.library.controller;

import com.library.model.Book;
import com.library.repository.BookRepository;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;

import java.util.List;

@RestController
@RequestMapping("/books")
public class BookController {

    @Autowired
    private BookRepository bookRepository;

    @GetMapping
    public List<Book> getAllBooks() {
        return bookRepository.findAll();
    }

    @PostMapping
    public Book createBook(@RequestBody Book book) {
        return bookRepository.save(book);
    }

    @GetMapping("/{id}")
    public Book getBookById(@PathVariable Long id) {
        return bookRepository.findById(id).orElse(null);
    }

    @PutMapping("/{id}")
    public Book updateBook(@PathVariable Long id, @RequestBody Book bookDetails) {
        Book book = bookRepository.findById(id).orElse(null);
        if (book != null) {
            book.setTitle(bookDetails.getTitle());
            book.setAuthor(bookDetails.getAuthor());
            return bookRepository.save(book);
        }
        return null;
    }

    @DeleteMapping("/{id}")
    public void deleteBook(@PathVariable Long id) {
        bookRepository.deleteById(id);
    }
}

package com.library;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication

```

```

public class LibraryManagementApplication {
    public static void main(String[] args) {
        SpringApplication.run(LibraryManagementApplication.class, args);
    }
}

```

Output:

```

demo - java - java --enable-native-access=ALL-UNNAMED -classpath ~/m2/wrapper/dists/apache-maven-3.9.10/e5402a93/boot/plexus-classworlds-2.9.0.jar -Dclassworlds.conf=/Users/svkeerthi/m2/wrapper/dist...
[INFO] Attaching agents: []

:: Spring Boot ::
(v3.5.3)

2025-07-04T13:12:25.970+05:30 INFO 63287 --- [demo] [main] librarymanagement.DemoAppl
ication : Starting DemoApplication using Java 21.0.1 with PID 63287 (/Users/svkeerthi/Desk
top/demo/target/classes started by svkeerthi in /Users/svkeerthi/Desktop/demo)
2025-07-04T13:12:25.971+05:30 INFO 63287 --- [demo] [main] librarymanagement.DemoAppl
ication : No active profile set, falling back to 1 default profile: "default"
2025-07-04T13:12:26.206+05:30 INFO 63287 --- [demo] [main] .s.d.r.c.RepositoryConfigu
rationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2025-07-04T13:12:26.223+05:30 INFO 63287 --- [demo] [main] .s.d.r.c.RepositoryConfigu
rationDelegate : Finished Spring Data repository scanning in 14 ms. Found 1 JPA repository interf
ace.
2025-07-04T13:12:26.372+05:30 INFO 63287 --- [demo] [main] o.s.b.w.embedded.tomcat.To
mcatWebServer : Tomcat initialized with port 8080 (http)
2025-07-04T13:12:26.379+05:30 INFO 63287 --- [demo] [main] o.apache.catalina.core.Sta

```

```

demo - java - java --enable-native-access=ALL-UNNAMED -classpath ~/m2/wrapper/dists/apache-maven-3.9.10/e5402a93/boot/plexus-classworlds-2.9.0.jar -Dclassworlds....
2025-07-04T15:58:16.908+05:30 INFO 65957 --- [demo] [main] org.hiber
nate.orm.connections.pooling : HHH10001005: Database info:
Database JDBC URL [Connecting through datasource 'HikariDataSource (Hika
riPool-1)']
Database driver: undefined/unknown
Database version: 2.3.232
Autocommit mode: undefined/unknown
Isolation level: undefined/unknown
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2025-07-04T15:58:17.195+05:30 INFO 65957 --- [demo] [main] o.h.e.t.j
.p.i.JtaPlatformInitiator : HHH000489: No JTA platform available (set 'hib
ernate.transaction.jta.platform' to enable JTA platform integration)
2025-07-04T15:58:17.206+05:30 INFO 65957 --- [demo] [main] j.LocalCo
ntainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persi
stence unit 'default'
2025-07-04T15:58:17.274+05:30 WARN 65957 --- [demo] [main] JpaBaseCo
nfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default.
Therefore, database queries may be performed during view rendering. Explicitly
configure spring.jpa.open-in-view to disable this warning
2025-07-04T15:58:17.401+05:30 INFO 65957 --- [demo] [main] o.s.b.w.e
mbedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with contex
t path '/'
2025-07-04T15:58:17.404+05:30 INFO 65957 --- [demo] [main] libraryma
nagement.DemoApplication : Started DemoApplication in 1.413 seconds (proc
ess running for 1.565)

```